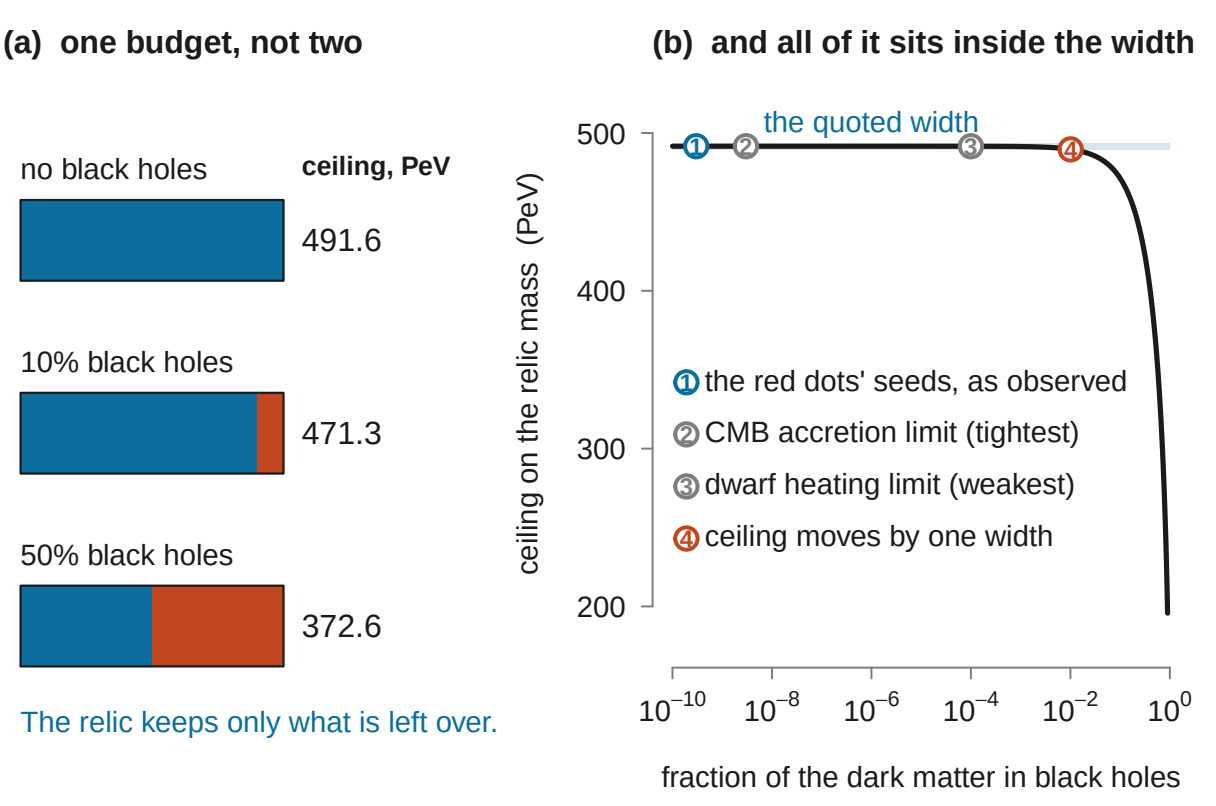
(a) one budget, not two
no black holes
ceiling, PeV
491.6
10% black holes
471.3
50% black holes
372.6
The relic keeps only what is left over.
(b) and all of it sits inside the width
the quoted width
ceiling on the relic mass (PeV)
500
400
300
200
① the red dots' seeds, as observed
② CMB accretion limit (tightest)
③ dwarf heating limit (weakest)
④ ceiling moves by one width
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
10^{0}
fraction of the dark matter in black holes